Standard mode 14:05

```
sublistă([], _).
sublistă([H|T], [H|L]) :- sublistă(T, L).
sublistă([H|T], [_|L]) :- sublistă([H|T], L).
sublistele(L, LS) :- setof(S, sublistă(S, L), LS).
surjectivă(R, B) :- not((member(Y, B), not(member((_, Y),
    R)))).
injectiva(R) :- not((member((A,B),R),
    member((X,B),R), A\=X)).
```

$R \subseteq A \times B \supseteq (R^{-1})^{-1}$

$R^{-1} \subseteq B \times A$

$\{(b,a) \mid \underbrace{(a,b) \in R}_{\text{Ilustr}}\}$

aRb

$(R^{-1})^{-1} = R$

$\forall x = (a,b) \in A \times B.$

$a(R^{-1})^{-1}b \Leftrightarrow bR^{-1}a \Leftrightarrow aRb$

Standard mode 14:05

```
im(R, A, Im) :- setof(Y, X^(member(X,A),
    member((X,Y),R)), Im), !.
im(_, _, []).

img([], _, []).
img([(X,Y)|T], A, [Y|U]) :- member(X,A), !,
    img(T, A, U).
img([_|T], A, U) :- img(T, A, U).

preim(R, B, PreIm) :- setof(X, Y^(member(Y,B),
    member((X,Y),R)), PreIm), !.
preim(_, _, []).

preim([], _, []).
preim([(X,Y)|T], B, [X|U]) :- member(Y,B), !,
    preim(T, B, U).
preim([_|T], B, U) :- preim(T, B, U).
```

Standard mode 14:05

```
egalmult(L,M) :- permutare(L,M).

stergeuna(_,[],_) :- fail.
stergeuna(H,[H|T],T).
stergeuna(X,[H|T],[H|U]) :- stergeuna(X,T,U).
?- stergeuna(a,[a,1,a,3,3,a],L).
?- stergeuna(a,DinCeLista,[1,3,3]).
reuniune(L,M,R) :- append(L,M,C),
                   elindupl(C,R).

reun(L,M,R) :- setof(X, member(X,L);member(X,M),R),!.
reun(_,_,[]).
elindupl([],[]).
elindupl([H|T],[H|L]) :- sterge(H,T,M), elindupl(M,L).
```

```
permutare([],[]).
permutare([H|T],P) :- permutare(T,Q),
                      stergeuna(H,P,Q).

permutarile(L,LP) :- setof(P,permutare(L,P),LP)
sterge(_,[],[]).
sterge(H,[H|T],L) :-
     sterge(H,T,L),!.
sterge(X,[H|T],[H|L]) :-
     sterge(X,T,L).
```

Standard mode 14:04

```
inters(L,M,I) :- setof(X,(member(X,L),member(X,M)),I)
                 ,!.
inters(_,_,[]).
intersectie([],_,[]).
intersectie([H|T],L,[H|U]) :- member(H,L),
                              !,intersectie(T,L,U).
intersectie([_|T],L,U) :- intersectie(T,L,U).
dif(L,M,D) :- setof(X,(member(X,L),
      not(member(X,M))),D),!.
dif(_,_,[]).
```

```
diferenta(L,[],L).
diferenta(L,[H|T],D) :- sterge(H,L,M),
                        diferenta(M,T,D).

xor(P,Q) :- P,not(Q);
            Q,not(P).
difsim(L,M,D) :- setof(X,xor(member(X,L),member(X,M)),D),!.
difsim(_,_,[]).
diferosim(L,M,D) :- diferenta(L,M,D1),
     diferenta(M,L,D2),reuniune(D1,D2,D).
intersect(L,M,I) :- intersectie(L,M,P), elindupl(P,I).

difer(L,M,D) :- diferenta(L,M,E), elindupl(E,D).
```

Exe. $T \in$ set, $A, B \in \mathcal{P}(T)$

$f : \mathcal{P}(T) \to \mathcal{P}(A) \times \mathcal{P}(B)$, $(\forall X \in \mathcal{P}(T))(f(X) = (X \cap A, X \cap B))$

(1) $f \to$ inj $\Leftrightarrow A \cup B = T$

(2) $f \to$ surj $\Leftrightarrow A \cap B = \emptyset$

(3) $f \to$ bij $\Leftrightarrow A = \overline{B} \Leftrightarrow B = \overline{A}$, unde $(\forall M \in \mathcal{P}(T))(\overline{M} = T \setminus M)$

$M = N \Rightarrow \overline{M} = \overline{N}$; $M = \overline{\overline{M}}$, $N = \overline{\overline{N}}$

$x \in \overline{M} \Leftrightarrow x \notin M$

(1),(2) $\Rightarrow$ $f \to$ bij $\Leftrightarrow \begin{cases} f \to \text{inj și} \\ f \to \text{surj} \end{cases} \overset{(1),(2)}{\Leftrightarrow} \begin{cases} A \cup B = T \\ A \cap B = \emptyset \end{cases} \Leftrightarrow A = \overline{B} \Leftrightarrow B = \overline{A}$

$\Leftrightarrow A \supseteq \overline{B}$ $(\Leftrightarrow B \supseteq \overline{A})$

$\Leftrightarrow A \subseteq \overline{B}$ $(\Leftrightarrow B \subseteq \overline{A})$

Fie $x \in T$.

$A = \overline{B} \Leftrightarrow \overline{A} = \overline{\overline{B}} \Leftrightarrow \overline{A} = B$

$\overline{\overline{B}} = B \Leftarrow x \in \overline{\overline{B}} \Leftrightarrow x \notin \overline{B} \Leftrightarrow \text{not}(x \in \overline{B}) \Leftrightarrow \text{not}(\text{not}(x \in B)) \Leftrightarrow x \in B$

$A \cup B = T \;/\; \cap \overline{B} \Rightarrow (A \cup B) \cap \overline{B} = \underbrace{T \cap \overline{B}}_{= \overline{B}}$

$(A \cap \overline{B}) \cup \underbrace{(B \cap \overline{B})}_{= \emptyset} = A \cap \overline{B} \subseteq A$

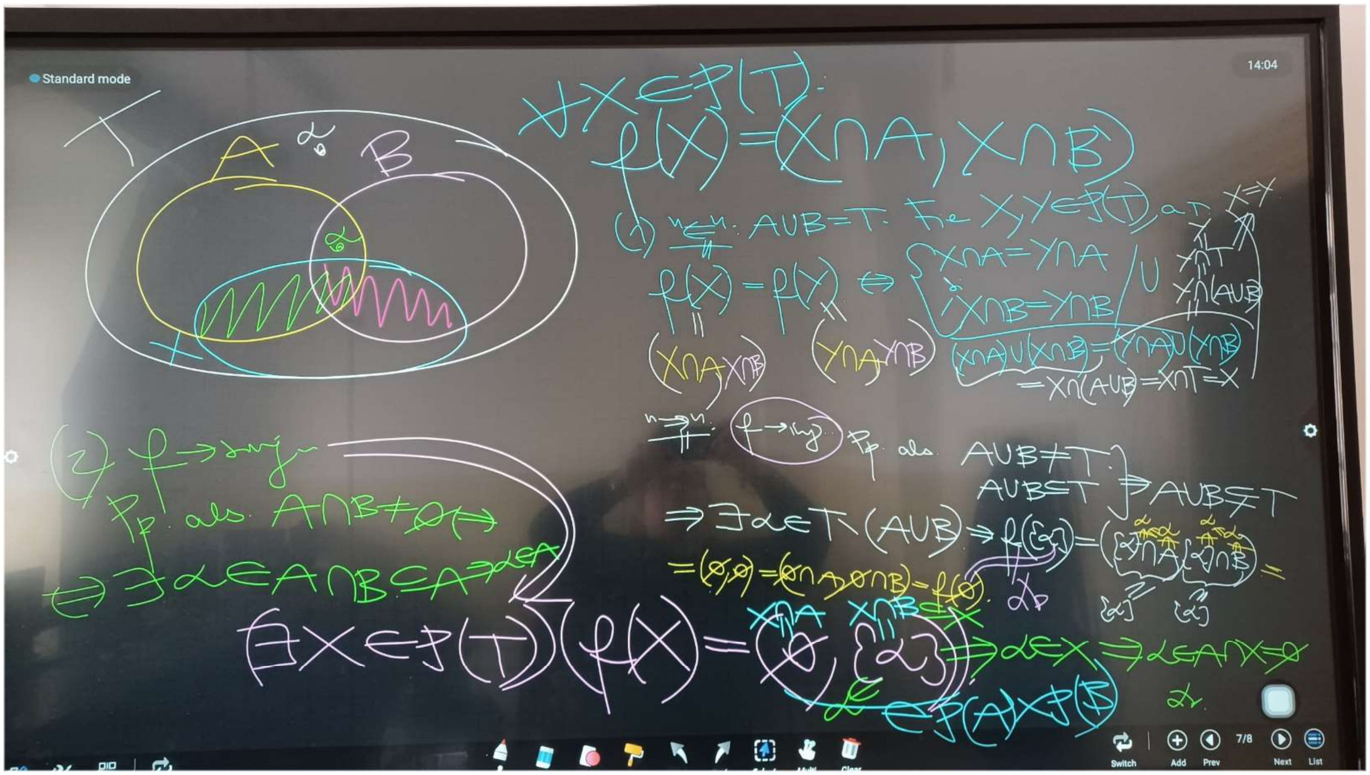
$\Rightarrow \overline{B} \subseteq A \;/\; \cup B \Rightarrow \underbrace{B \cup \overline{B}}_{= T} \subseteq \underbrace{A \cup B}_{\subseteq T} \Rightarrow A \cup B = T$

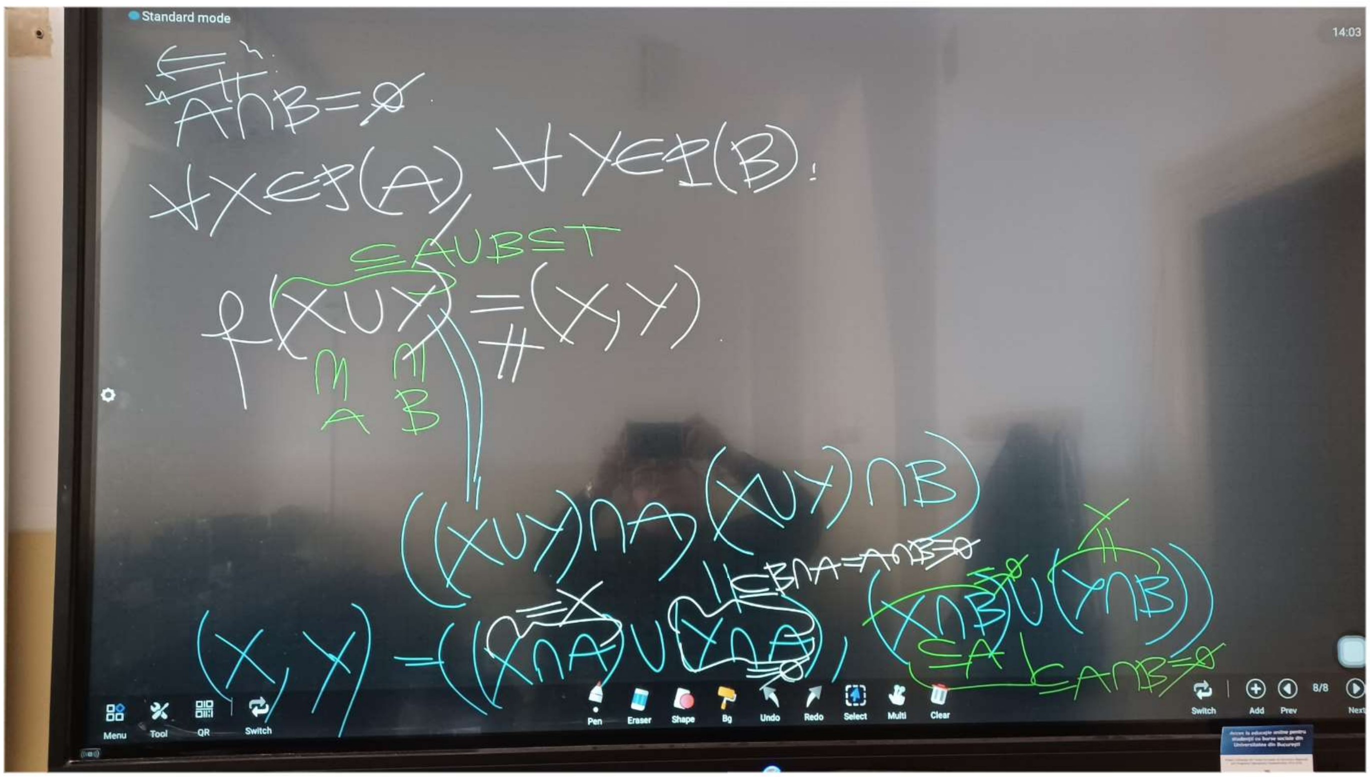
$A \cap B = \emptyset \;/\; \cup \overline{B} \Rightarrow (A \cap B) \cup \overline{B} = \overline{B} \Rightarrow \overline{B} \supseteq A$

$(A \cup \overline{B}) \cap \underbrace{(B \cup \overline{B})}_{= T} = A \cup \overline{B} \supseteq A$

$\Rightarrow \left(\underset{B \cup A}{A \cup B} = T \Leftrightarrow \overline{B} \subseteq A \Leftrightarrow \overline{A} \subseteq B \right)$

$\Rightarrow \left(\underset{B \cap A}{A \cap B} = \emptyset \Leftrightarrow A \subseteq \overline{B} \Leftrightarrow B \subseteq \overline{A} \right)$


Standard mode
14:04
∀X∈P(T):
f(X)=(X∩A, X∩B)
AUB=T
f(X)=f(Y)
(X∩A)∪(X∩B)=(Y∩A)∪(Y∩B)
=X∩(AUB)=X∩T=X
AUB≠T
A∩B≠∅
∃X∈P(T)
7/8
Switch
Add
Prev
Next
List


Standard mode
14:03
A∩B=∅
∀X∈P(A) ∀Y∈P(B):
⊆AUB⊆T
f(X∪Y)=(X,Y)
((X∪Y)∩A, (X∪Y)∩B)
⊆A
⊆A∩B=∅
Menu
Tool
QR
Switch
Pen
Eraser
Shape
Bg
Undo
Redo
Select
Multi
Clear
Add
Prev
8/8
Next